Lab 6 Rubric: Centripetal Force

Out of 100 points. This is the rubric on the LB6-M5 assignment in Brightspace. Lab 6 page · Word copy · PDF copy

Use this rubric as your writing guide: the criteria follow the lab report template top to bottom. Full-credit descriptors say exactly what a complete answer contains, so read them before you write rather than after.

Grammar, spelling, and formatting problems are assessed 1 point per occurrence, capped at 3 points for the report. A fault already scored under a criterion is not deducted again as a formatting occurrence. Deadline: a report submitted by the deadline may be revised and resubmitted up to two more times. A report that arrives after the deadline but before grades are posted is graded as on time but cannot be revised; one that arrives after grades are posted is late, may take a penalty, and gets no detailed feedback.

The phrase “human error” earns no credit anywhere in the report. A percent difference under 10% between the mass from your slope and the mass from the balance may be attributed to ordinary experimental uncertainty, provided you say so explicitly and name a mechanism. The intercept is not noise — it is evidence, and a report that ignores it loses marks in Discussion.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Heading	3	**3** Names, course and section, lab title and date, all present and correctly formatted. No template placeholders.	**2** One item missing or misformatted.	**2** Template placeholders left in the heading block.	**1** Several items missing.	**0** No heading block.
Abstract	7	**7** Four sentences or fewer. States what was measured, gives the stopper mass from the graph with its uncertainty and the percent difference against the balance, and states whether the model held. No methods.	**6** All content present but runs long or includes method detail.	**4** Reports a result but omits the uncertainty or the comparison.	**2** Describes the lab but reports no numbers.	**0** No abstract.
Introduction	8	**8** Builds $F_c = (m/r)v^2$ line by line from $T = M_h g$ and $F_c = mv^2/r$, with every symbol defined, and states what the slope must physically be. Ends with a testable hypothesis.	**7** All elements present; one step or definition thin.	**5** The relation is quoted rather than built, or the slope is never interpreted.	**3** Equation given with no derivation and no definitions.	**0** No introduction.
Experimental: Materials & Setup	8	**8** Apparatus described in prose; a labelled sketch with the radius marked; masses of stopper, washers AND paper clip given. States what is held constant.	**7** Sketch and masses present; the paper clip or the constant-radius statement missing.	**5** Sketch present but unlabelled, or a mass missing.	**3** Bare equipment list, or no sketch.	**0** No materials or setup description.
Experimental: Procedure	8	**8** Written in own words as prose, reproducible. Explains how the radius was held constant and why 30 revolutions were timed rather than one.	**7** Reproducible and in own words; the 30-revolution reasoning is thin.	**5** Reproducible, but the constant-radius method is absent.	**3** Copied or closely paraphrased from the handout.	**0** No procedure.
Results: Data Tables	12	**12** Table of all five runs carrying washers, hanging mass, F_c, time for 30 rev, period, v and v^2. Captions above, units in header rows, consistent significant figures.	**10** Table present; minor issues with captions, units or significant figures.	**7** Table present but missing the period or v^2 column.	**4** Fewer than five runs tabulated.	**0** No data tables.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Results: Graphs	**8**	**8** F_c against v^2 with a fitted straight line, axes labelled with units, caption below. Both fits reported or the plotted one clearly identified.	**7** Graph present and correctly plotted; caption or axis labels incomplete.	**5** Axes transposed, so the slope is r/m rather than m/r, but otherwise correct.	**3** Points joined dot-to-dot, or no fitted line.	**0** No graph.
Results: Sample Calculations	**13**	**13** One worked sample of each type with units at every step and numbers shown going in: force, period, speed, speed squared, mass from slope, percent difference.	**11** All types present with minor unit or rounding slips.	**8** Three or four of the six types worked.	**4** Some types missing, or units only in the final answer.	**0** No sample calculations.
Discussion	**15**	**15** Opens with the mass from the slope and its percent difference, then argues it. Addresses the intercept by name, accounts for the DIRECTION of the offset, tests a physical explanation against the data, and chooses between the two mass values with a reason.	**12** Most of the above; one element thin, usually the direction of the offset.	**9** Interprets the result but treats the intercept as noise rather than evidence.	**5** Restates the results without interpreting them.	**0** No discussion.
Additional Questions	**8**	**8** Three of the five chosen and named, each answered fully in a dedicated section, in paragraphs with working shown and not restating the question.	**6** Three answered but one thinly, or the three chosen are not named.	**4** Answered inside another section, or only two answered.	**2** Answered as a bare list, or fewer than two answered.	**0** Not answered.
Conclusion	**5**	**5** Three to four sentences: numerical result, definitive verdict, one concrete improvement.	**4** Present but missing the numbers or the verdict.	**3** Present with numbers but no verdict.	**2** Vague, or a repeat of the discussion.	**0** No conclusion.
References	**5**	**5** At least one source beyond the textbook; Note 1 and the handout count. Author, title and date, consistently formatted.	**4** Multiple references but formatting incomplete.	**3** Textbook and handout only, properly formatted.	**2** Handout only.	**0** No references.
Total	**100**					

Overall: Complete 90+ · Nearly there 75+ · Needs work 50+ · Missing below 50.

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